Introduction to Quantum Physics and Structure of Matter

Lecture 4

October 24, 2026

TU/e EINDHOVEN UNIVERSITY OF TECHNOLOGY

https://docs.google.com/forms/d/13kKrkLGpWBiuvfKssoIrRy0I9PtTEOk_me6Fg2ZXxHA/edit#responses

Course Contents

Lecture	Date	Topics	Material
1	03-09-2025	•Relativistic dynamics •Review electromagnetic waves •Photoelectric effect •Blackbody radiation	Book: Ch 2.1-2.5, 2.7, 3.1-3.3
2	10-09-2025	•Compton scattering •De Broglie's hypothesis •Experimental evidence for De Broglie waves •Uncertainty relationships for classical particles •Heisenberg uncertainty relationships	Book: Ch. 3.4, 4.1-4.4
3	17-09-2025	•Behaviour of a wave at a boundary •Confining a particle •The Schrödinger equation •Applications of the Schrödinger equation	Book: Ch. 5.1-5.4
4	**24-09-2025**	**•Basic properties of atoms** **•Scattering experiments and the Thomson model** **•The Rutherford nuclear atom** **•Line spectra** **•The Bohr model** **•The Franck-Hertz experiment** **•Deficiencies of the Bohr model**	**Book: Ch. 6.1-6.6, 6.8**

Course Contents

Lecture	Date	Topics	Material
5	01-10-2025	•A one-dimensional atom •Angular momentum in the hydrogen atom •The hydrogen atom wave functions •Radial probability densities •Angular probability densities	Book: Ch. 7.1-7.7
6	08-10-2025	•The Pauli exclusion principle •Electron states in many-electron atoms •Outer electrons: screening and optical transitions •Inner electrons: absorption edges and X-rays	Book: Ch. 8.1-8.3, 8.5
7	15-10-2025	•Nuclear constituents •Nuclear sizes and shapes •Alpha decay •Beta decay •Gamma decay and nuclear excited states	Book: Ch. 12.1-12.3 12.6-12.9
8	t.b.d.	Visit to the Catharina hospital, radiotherapy department	
Final Exam: Thursday October 30th, 13:30-16:30			

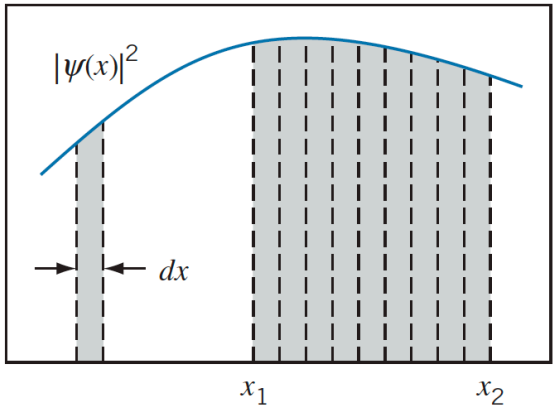
Recap Previous Lecture

- Schrödinger's Equation

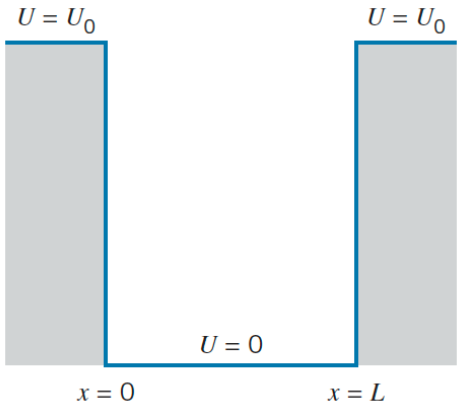
$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2}\right) + U\psi$$

- Probabilities

$$P(x)dx = |\psi(x)|^2 dx$$



- Potential Energy Wells



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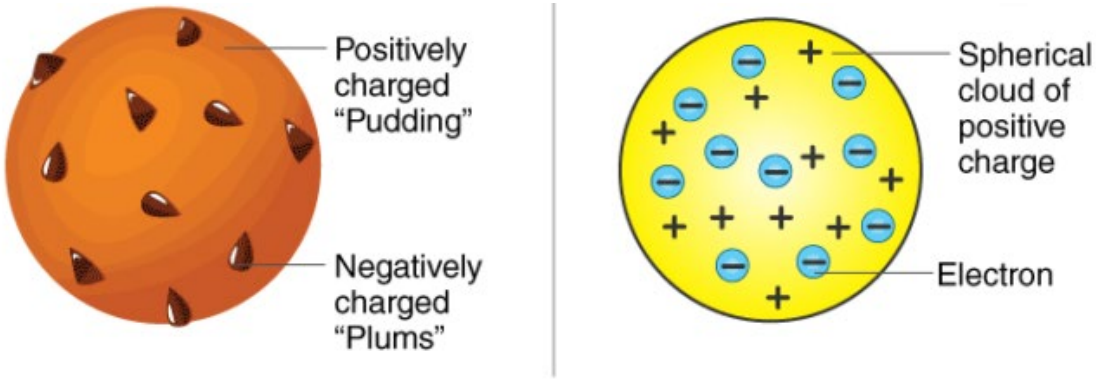
Atoms

- Already known in 1913:

1. Atoms are very small: 1000x smaller than visible light ($\sim 0.1\ \mathrm{nm}$)
2. Atoms are stable: they don’t spontaneously break apart
3. Atoms contain negatively charged electrons, but are electrically neutral
4. Atoms emit and absorb electromagnetic radiation

Atom Model – Thomson

- "Plum Pudding" model
- Z electrons, uniformly distributed



Jospeh John Thomson
1856-1940



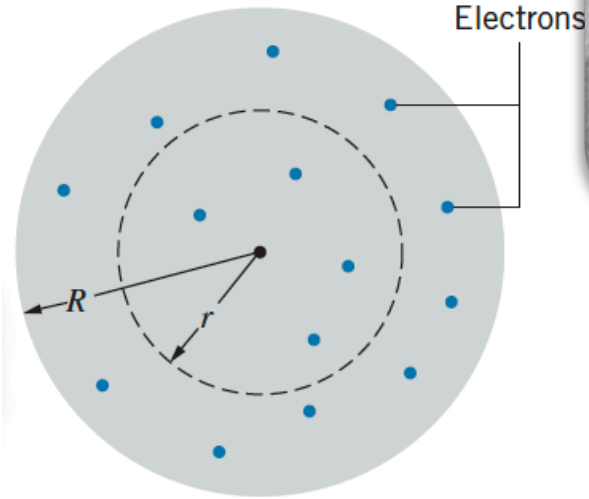
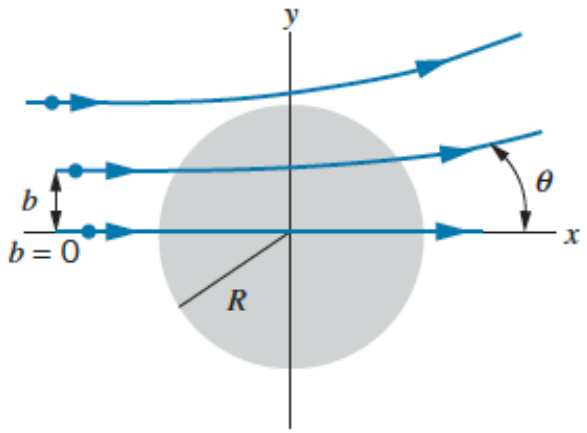


FIGURE 6.1 The Thomson model of the atom. Z electrons are imbedded in a uniform sphere of positive charge Ze and radius R. An imaginary spherical surface of radius r contains a fraction r^3/R^3 of the positive charge.

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Atom Model – Thomson

- Experimental verification with a *scattering experiment*
- Deflection occurs due to electrical forces
- Single atom impossible, so we use a thin foil



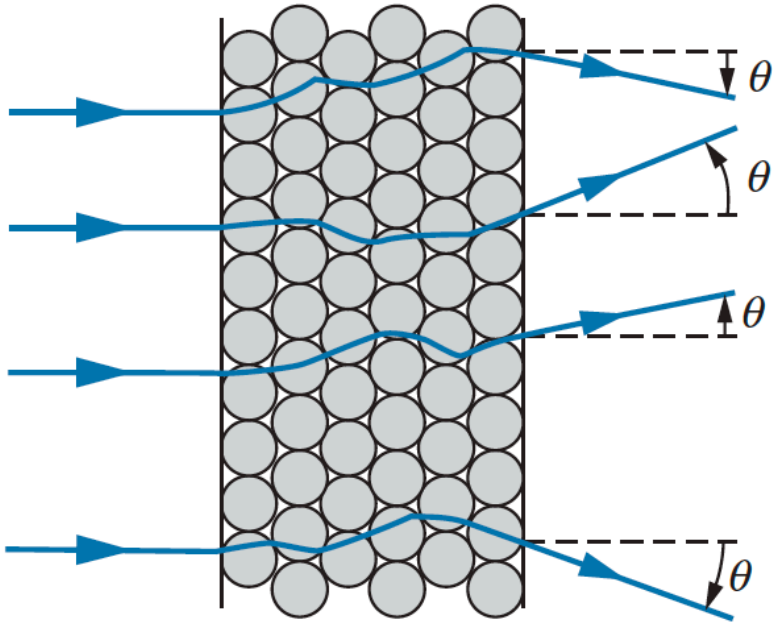
Foil Thickness: $1\ \mu m\ \approx 10^4\ atoms$

$$\theta = \sqrt{N}\theta_{avg}$$

Thomson model: $\theta_{avg} = 0.01°$

So: $\theta \approx 1°$

Experiments? ✓

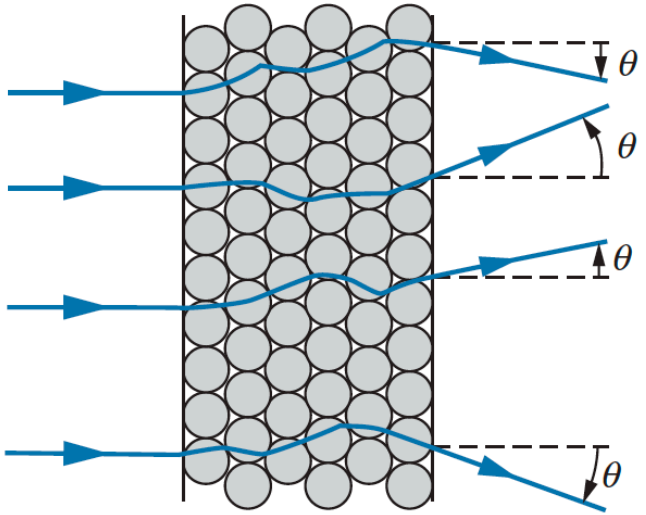


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Atom Model – Thomson

- Thomson's Model fails when looking at the probability of large scattering angles

- Each interaction deflects the projectile around $0.01°$
- To observe a $90°$ deflection → about 10^4 successive scatterings in the same direction



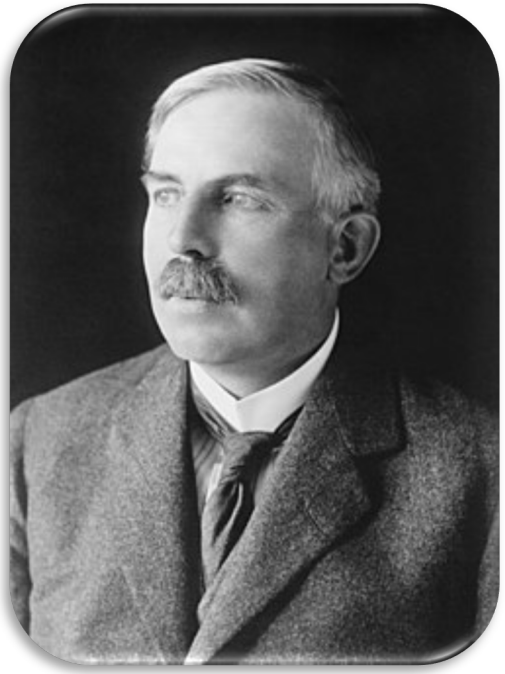
- 50/50 chance for up/down scattering:

$$\left(\frac{1}{2}\right)^{10{,}000} = 10^{-3000}$$

- Experiments shows that the probability was: 10^{-4}

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Ernest Rutherford
1871-1937



Atom Model – Rutherford

- *"It was quite the most incredible event that ever happened to me in my life. It was as incredible as if you fired a 15-inch shell at a piece of tissue paper and it came back and hit you."*

- Positive charge is concentrated in an extremely small region of 10^{-14} m

Atom Model – Rutherford

- Rutherford model explains higher probability of large θ's
- A projectile of charge ze experiences a repulsive force:

$$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1||q_2|}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{(ze)(Ze)}{r^2}$$

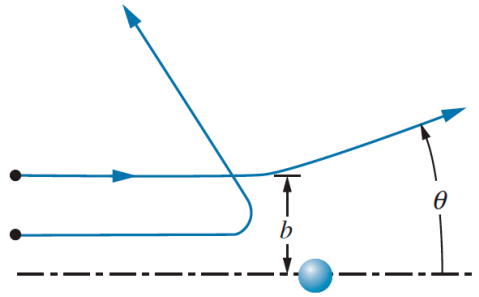


FIGURE 6.4 Scattering by a nuclear atom. The path of the scattered particle is a hyperbola. Smaller impact parameters give larger scattering angles.

$$\frac{1}{r} = \frac{1}{b}\sin\phi + \frac{zZe^2}{8\pi\epsilon_0 b^2 K}(\cos\phi - 1)$$

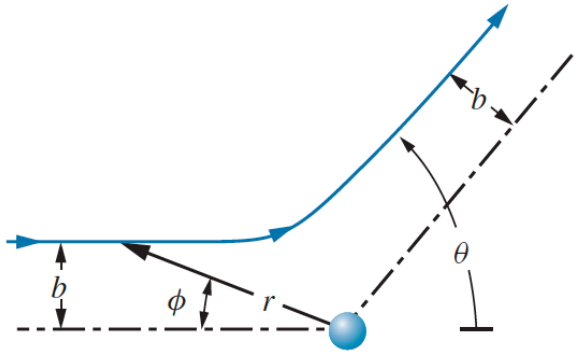
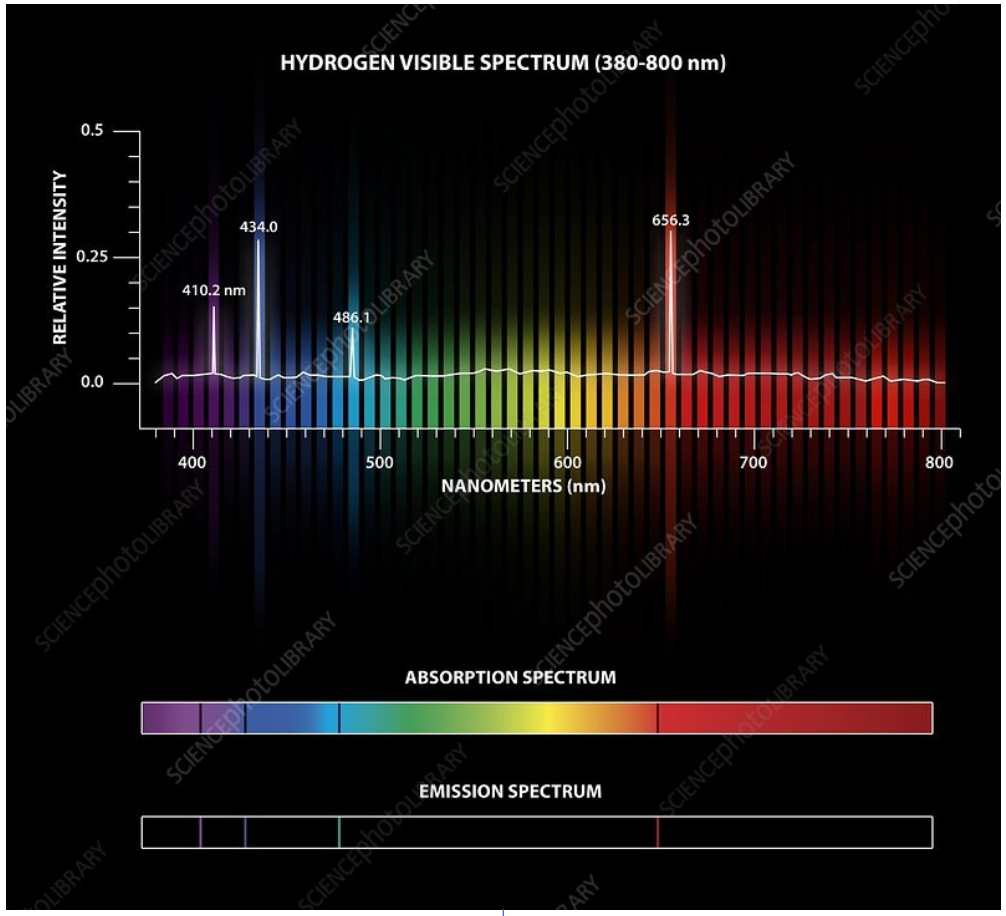


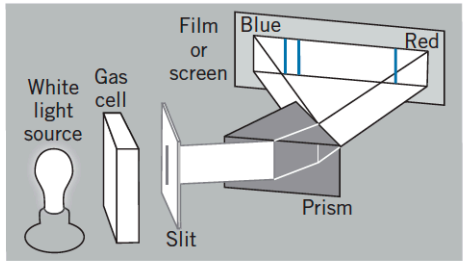
FIGURE 6.5 The hyperbolic trajectory of a scattered particle.

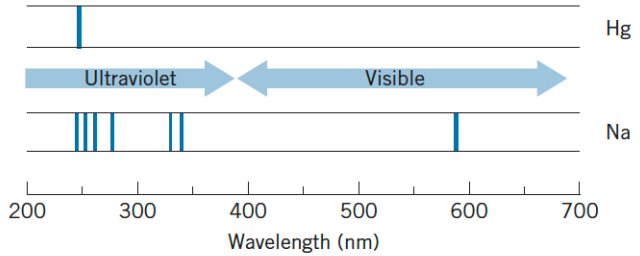
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Emission & Absorption

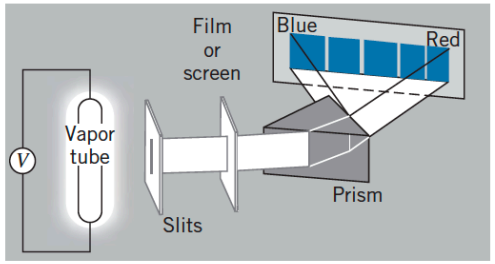


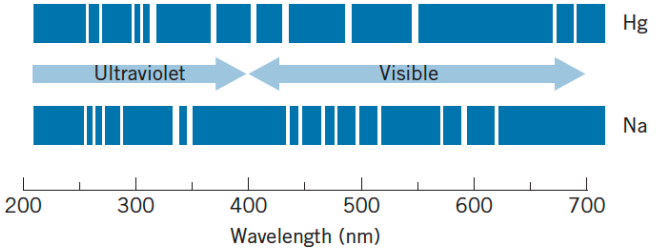
Absorption spectra





Emission spectra





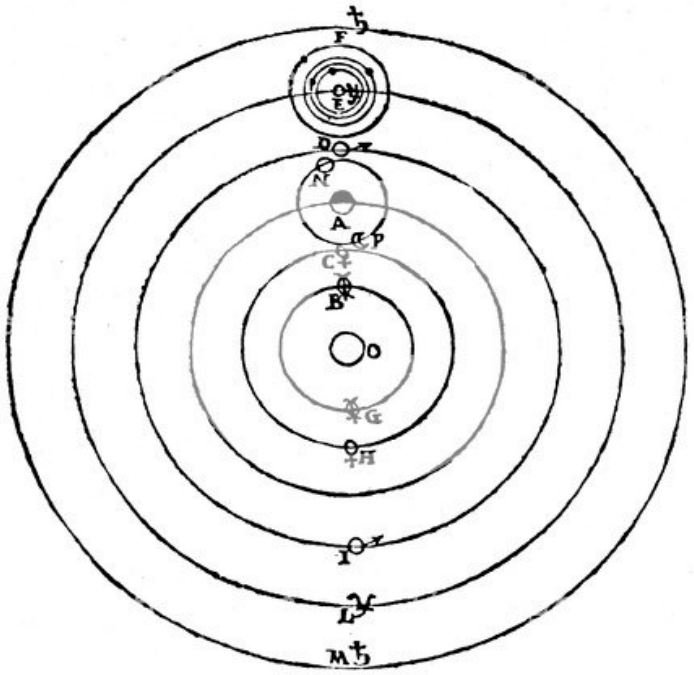
Balmer (1885): $\lambda = 364.5\ nm\ \frac{n^2}{(n^2-4)}$ $(n = 3,4,5, \ldots;\ \lambda = 656.1, 486, 434, 410, \ldots)$

Atom Model – Bohr

Niels Bohr
1885-1962



- ”Atoms resemble a miniature planetary system”



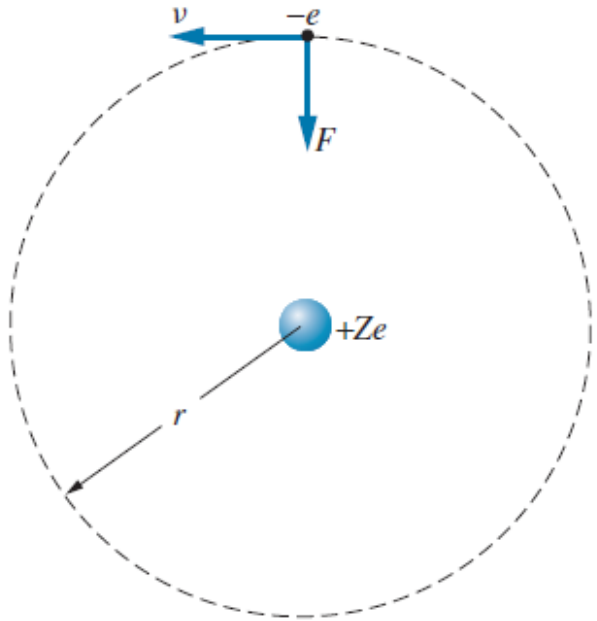


FIGURE 6.17 The Bohr model of the atom ($Z = 1$ for hydrogen).

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Atom Model – Bohr

- The atom doesn't collapse under the influence of the electrostatic Coulomb force of the nucleus on the electrons
- Consider a hydrogen atom:

$$F_{el} = F_{cp}$$

$$F_{cp} = ma_{cp} = m\frac{v^2}{r}$$

$$F_{el} = F_{cp} \Rightarrow \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} = mv^2$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

Potential energy on charge q:

$$U = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r} = \frac{-1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

$$E = K + U$$

$$E = \frac{1}{2}\frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} - \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r}$$

$$E = -\frac{1}{8\pi\epsilon_0}\frac{e^2}{r}$$

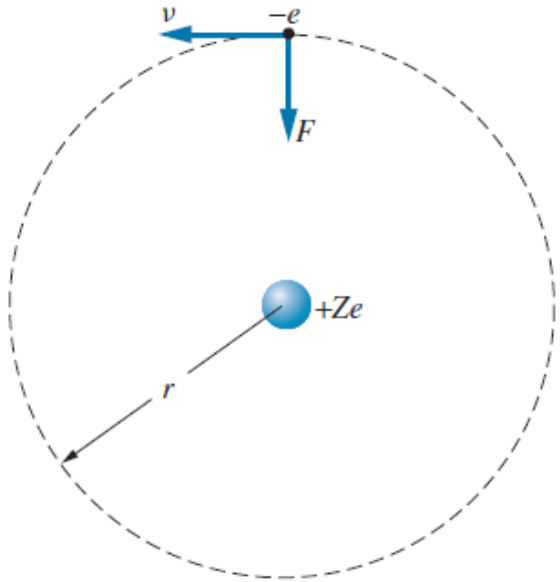
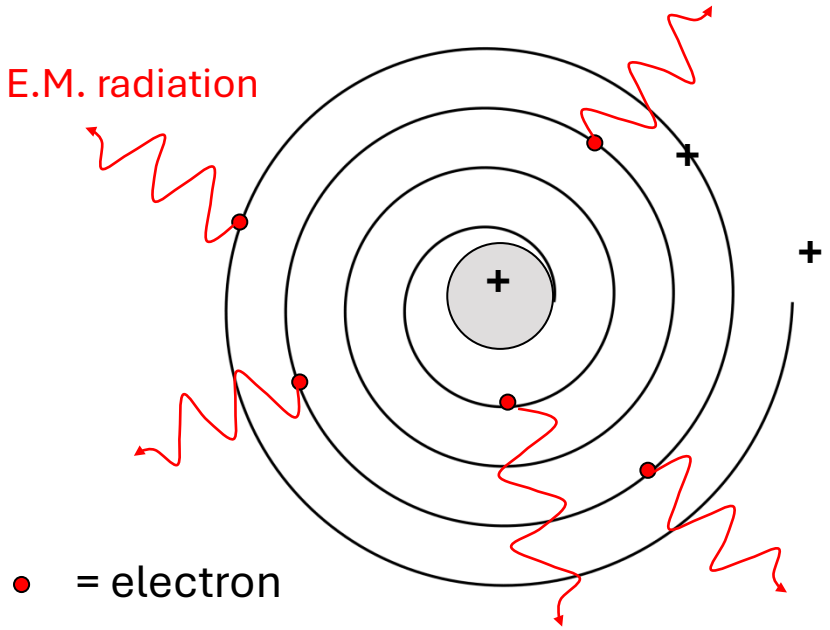


FIGURE 6.17 The Bohr model of the atom ($Z = 1$ for hydrogen).

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Atom Model – Bohr

- So, everything is solved now?



Classical EM theory: accelerated electron (as in H-atom) produces EM radiation, hence it loses energy and will go in a “spiral trajectory” to nucleus.

But: this does not happen!

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Atom Model – Bohr

Niels Bohr
1885-1962



- Bohr's hypothesis to overcome this issue:

1. *Stationary states* exists in which the electron may exist without radiating electromagnetic energy

2. Energy is absorbed or radiated in discrete amounts $E = hf$ only when an electron switches from one state to another*, which results in quantized
   angular momentum $\vec{L} = \vec{r} \times \vec{p} = n\hbar \ \ (n = 1,2,3, \dots)$

*) In 1925, De Broglie reinterpreted this theorem by stating that in a stationary orbit the electron's De Broglie wave must be a standing wave around a circular orbit: $n\lambda = 2\pi r$; with $\lambda = h/p = h/mv$ it follows that $mvr = n\hbar$

Atom Model – Electron Orbit

- Angular momentum in a circular orbit: $L = m_e v r$
- Angular momentum is quantized (Bohr): $L = n\hbar$

$$K = \frac{1}{2} m_e v^2 = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r}$$

$$\frac{1}{2} m_e \left(\frac{n\hbar}{m_e r} \right)^2 = \frac{1}{8\pi\epsilon_0} \frac{e^2}{r}$$

Solving for r gives:

$$r_n = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} n^2 = a_0 n^2$$

With $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2} = 0.0529\ nm$ (Bohr Radius), and $n = 1,2,3, \ldots$

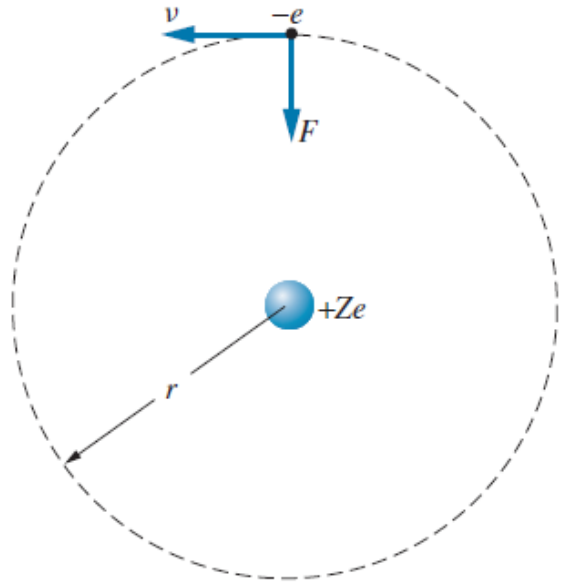
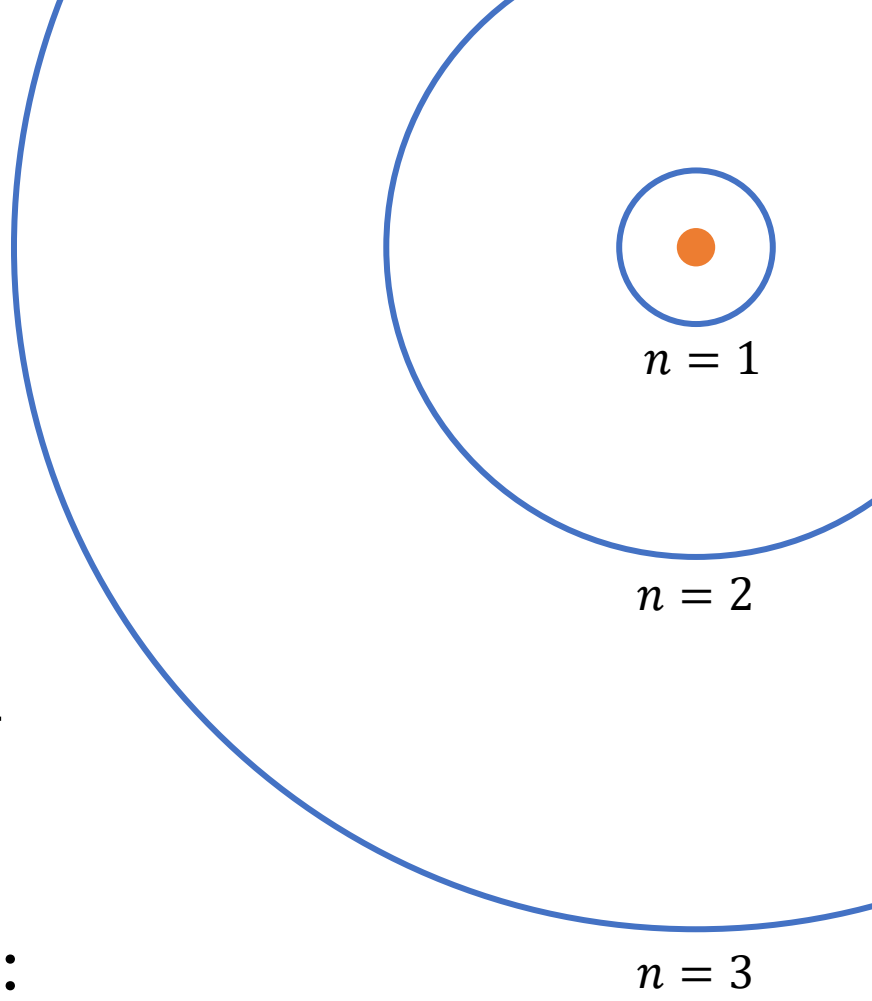


FIGURE 6.17 The Bohr model of the atom ($Z = 1$ for hydrogen).

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Atom Model – Electron Orbit

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} n^2 = a_0 n^2$$



- Important conclusions:
  - Only certain orbits are allowed by the Bohr model
  - Radii grow rapidly: $a_0, 4a_0, 9a_0, \ldots$

- We can calculate the corresponding energies:

$$E = -\frac{1}{8\pi\epsilon_0}\frac{e^2}{r} \quad \text{and} \quad r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} n^2$$

$$E_n = -\frac{m_e e^4}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2} = -\frac{A}{n^2} = -\frac{13.60\ \text{eV}}{n^2}$$

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Atom Model – Exercise

In the Bohr model of the hydrogen atom, the electron can only be found at discrete radial distances from the nucleus. Given that the radius for the ground state is $R = 5.29 \cdot 10^{-11}$ m, what is the radius for the third level?

Atom Model – Exercise

In the Bohr model of the hydrogen atom, the electron can only be found at discrete radial distances from the nucleus. Given that the radius for the ground state is $R = 5.29 \cdot 10^{-11}$ m, what is the radius for the third level?

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2} n^2 = a_0 n^2$$

$$r_1 = a_0 = 5.29 \cdot 10^{-11}\ \text{m}$$

$$r_3 = a_0 3^2 = 9a_0 = 9 \cdot (5.29 \cdot 10^{-11}\ \text{m}) = 4.76 \cdot 10^{-10}\ \text{m}$$

Atom Model – Electron Orbit

$$E_n = -\frac{m_e e^4}{32\pi^2 \epsilon_0^2 \hbar^2}\frac{1}{n^2} = -\frac{A}{n^2} = -\frac{13.60\ \text{eV}}{n^2}$$

- Quantized energy levels; for n= 1 (*ground state*): $E_1 = -13.6$ eV
- Photon emission during Transition from state n_1 to state n_2 ($n_1 > n_2$):

Given a photon with energy $h\nu = h\frac{c}{\lambda} = E_{n_1} - E_{n_2} = -\frac{A}{n_1^2} + \frac{A}{n_2^2} = A\ \frac{n_1^2 - n_2^2}{n_1^2 n_2^2}$

$$\lambda = \frac{hc}{A}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} = \frac{1}{R_\infty}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2}$$

With $R_\infty = \frac{A}{hc} = \frac{m_e e^4}{64\pi^3 \varepsilon_0^2 \hbar^3 c} = 1.097 \quad 10^7\,\text{m}^{-1}$ (The Rydberg constant)

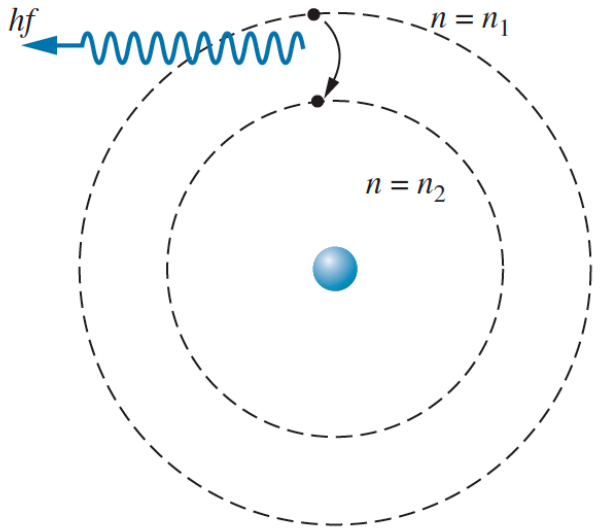


FIGURE 6.19 An electron jumps from the state n_1 to the state n_2 as a photon is emitted.

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Atom Model – Exercise

Find the wavelengths of the transitions from $n_1 = 3$ to $n_2 = 2$ and from $n_1 = 4$ to $n_2 = 2$ in atomic hydrogen

$$\lambda = \frac{hc}{A}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} = \frac{1}{R_\infty}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2}$$

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Atom Model – Exercise

Find the wavelengths of the transitions from $n_1 = 3$ to $n_2 = 2$ and from $n_1 = 4$ to $n_2 = 2$ in atomic hydrogen

$$\lambda_{n_1=3\to n_2=2} = \frac{1}{R_\infty}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} = \frac{1}{1.097 \quad 10^7\,\mathrm{m}^{-1}}\frac{3^2 2^2}{3^2 - 2^2} = 656.1\ \mathrm{nm}$$

$$\lambda_{n_1=4\to n_2=2} = \frac{1}{R_\infty}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} = \frac{1}{1.097 \quad 10^7\,\mathrm{m}^{-1}}\frac{4^2 2^2}{4^2 - 2^2} = 486.0\ \mathrm{nm}$$

$$\lambda = \frac{hc}{A}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2} = \frac{1}{R_\infty}\frac{n_1^2 n_2^2}{n_1^2 - n_2^2}$$

With $R_\infty = \frac{A}{hc} = \frac{m_e e^4}{64\pi^3 \varepsilon_0^2 \hbar^3 c} = 1.097 \quad 10^7\,\mathrm{m}^{-1}$

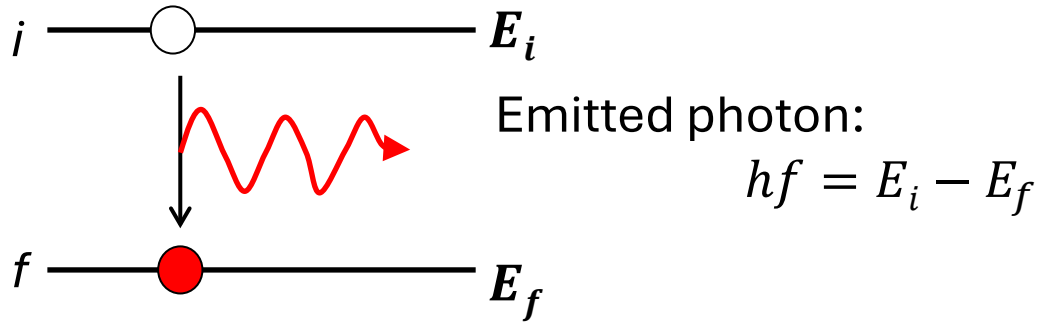
(The Rydberg constant)

Emission & Absorption

- **Emission**

electron goes from level *i* to **lower** level *f*

Energy difference is emitted as photon



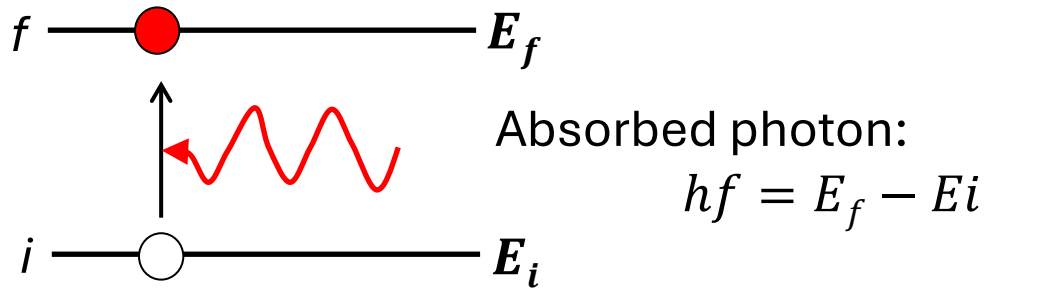
Emitted photon:

$$hf = E_i - E_f$$

- **Absorption**

Electron goes from level *i* to **highe**r level *f*

Energy difference is absorbed from photon



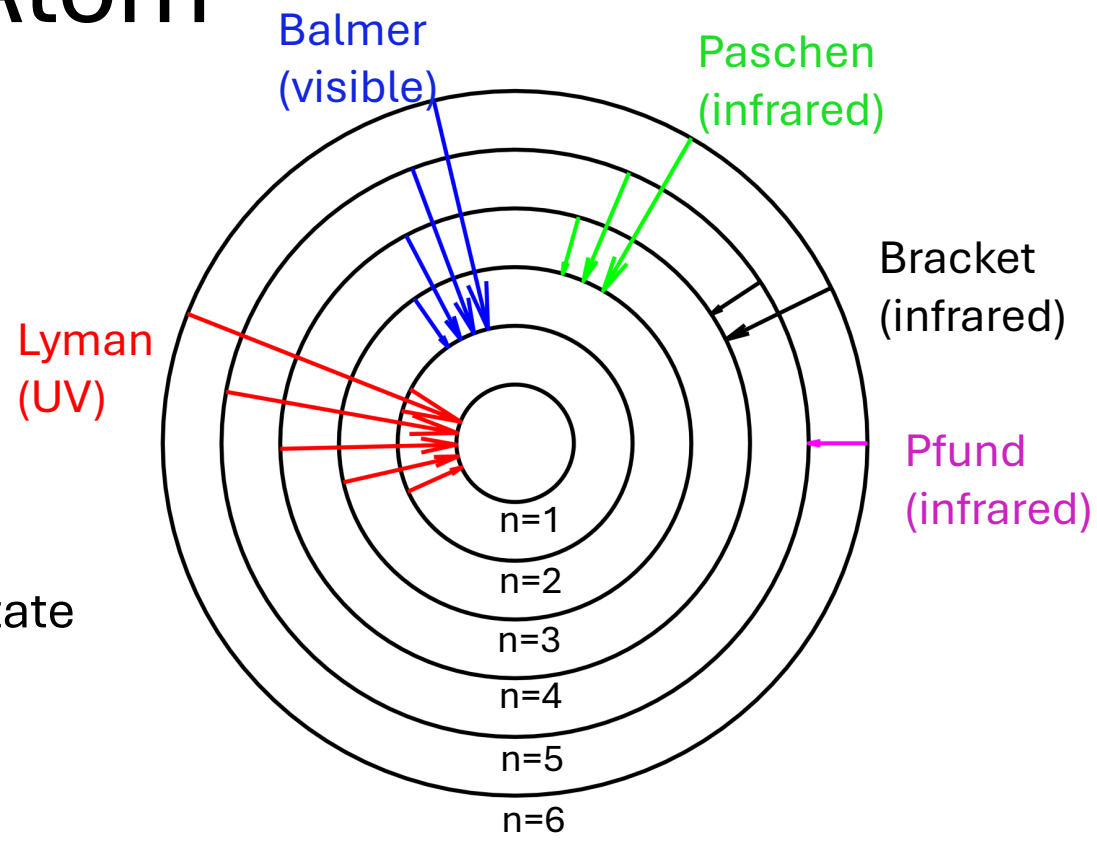
Absorbed photon:

$$hf = E_f - Ei$$

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Emission Lines Hydrogen Atom

$$E_n = -\frac{13.60\ \text{eV}}{n^2}$$

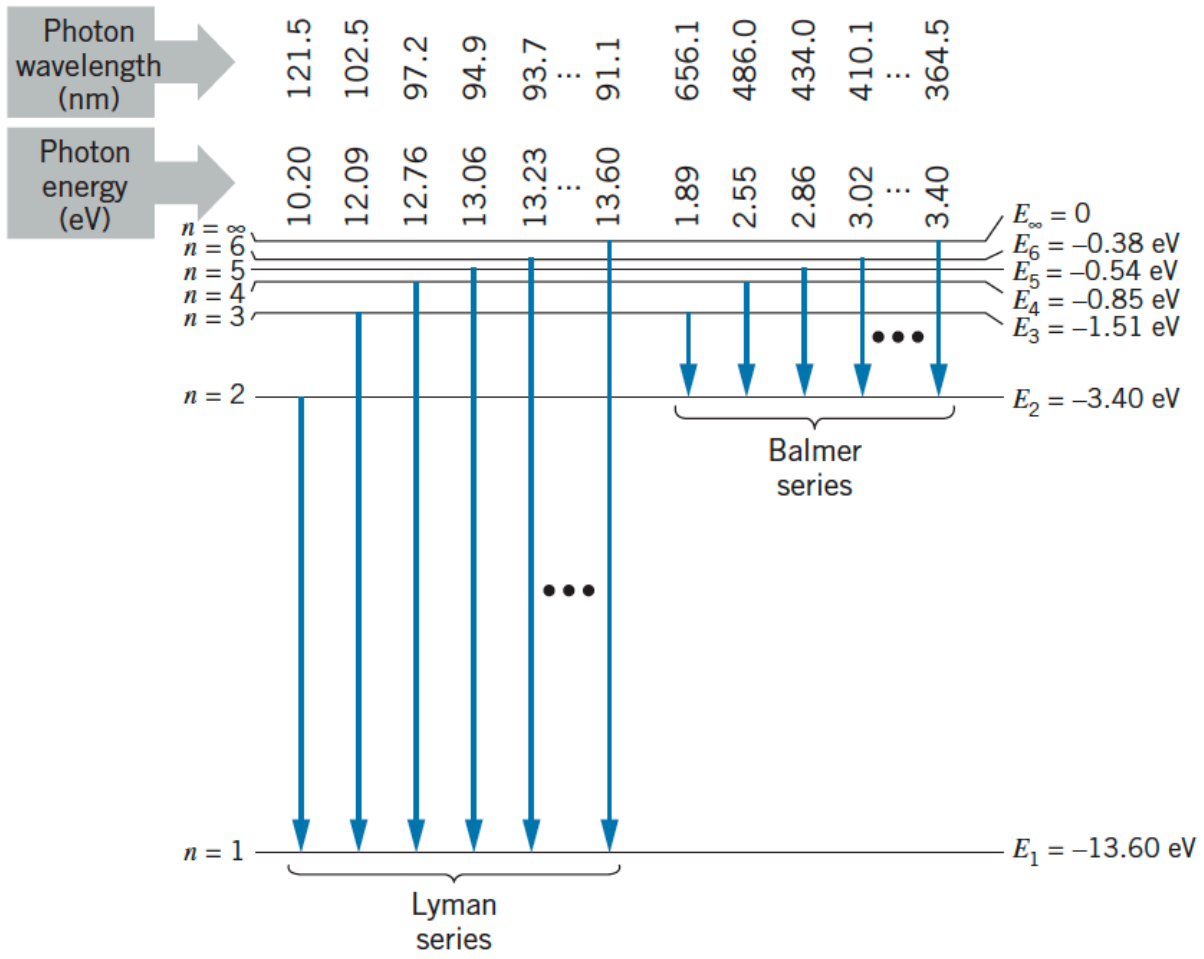


- At $r = \infty$, the potential is 0
- In level n the energy is negative; electron in bound state

Ground state: state with lowest energy ($E_1 = -13.6$ eV)
Excited state: states with higher energy ($n > 1$)
Ionization energy: energy needed to release the electron from the atom
Binding energy: magnitude of an electron's energy $|E_n|$

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Energy Diagram



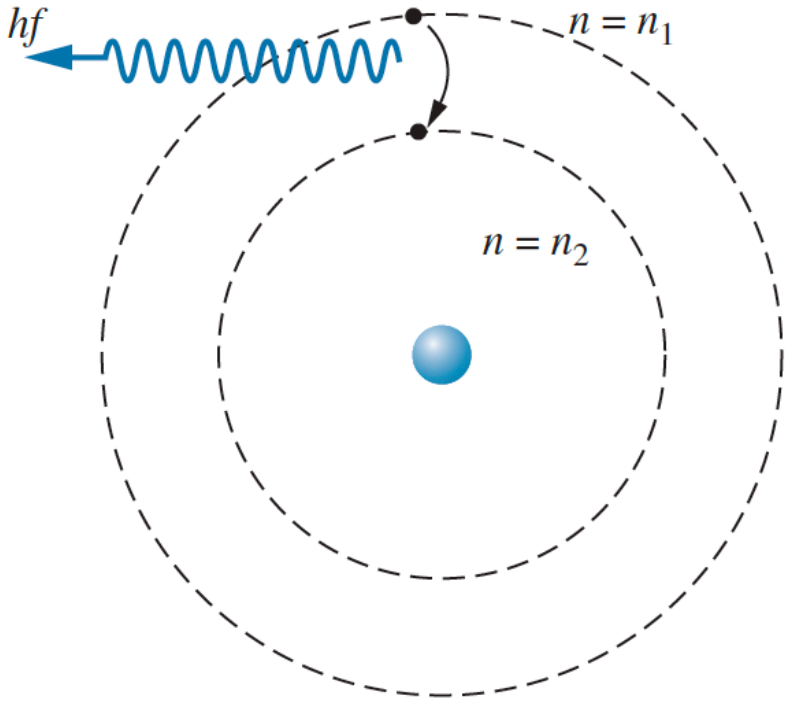


FIGURE 6.19 An electron jumps from the state n_1 to the state n_2 as a photon is emitted.

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Emission (& Absorption) Lines

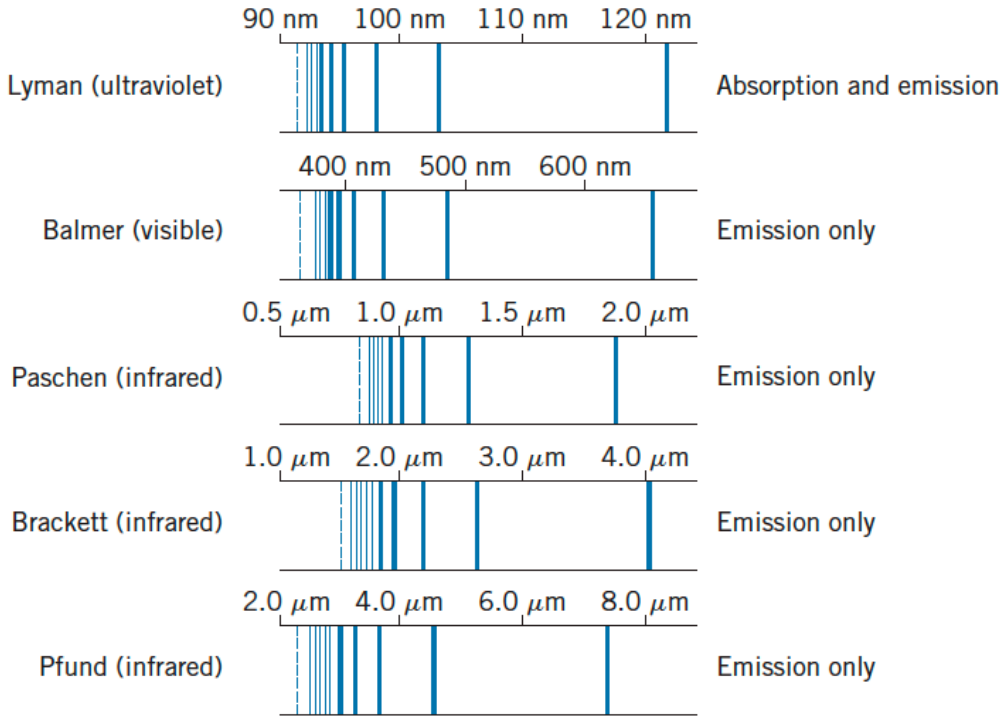
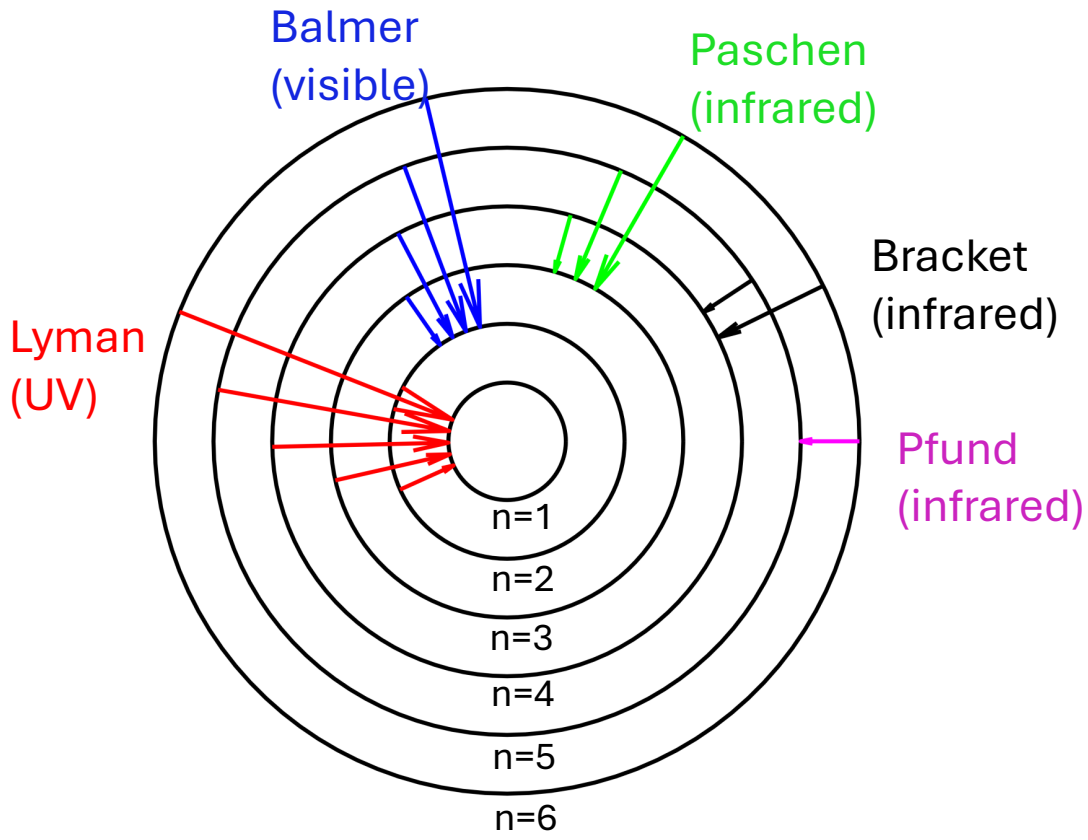
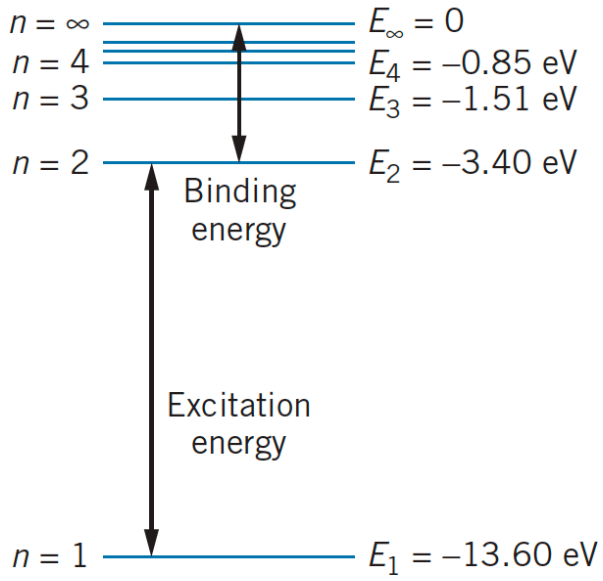


FIGURE 6.16 Emission and absorption spectral series of hydrogen. Note the regularities in the spacing of the spectral lines. The lines get closer together as the limit of each series (dashed line) is approached. Only the Lyman series appears in the absorption spectrum; all series are present in the emission spectrum.



Emission Lines – Exercise

An electron in a hydrogen atom has a binding energy of $0.85\ \mathrm{eV}$. It makes a transition to a state with an excitation energy of $10.2\ \mathrm{eV}$. Find the energy of the emitted photon in eV.



Emission Lines – Exercise

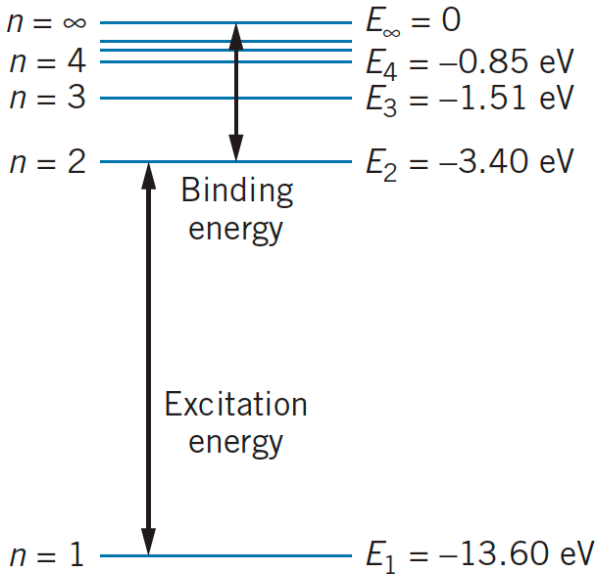
An electron in a hydrogen atom has a binding energy of 0.85 eV. It makes a transition to a state with an excitation energy of 10.2 eV. Find the energy of the emitted photon in eV.

Electron start in level $n = 4$: $|E_4| = 0.85$ eV

Excitation energy of 10.2 eV corresponds to the gap between $n = 1$ and $n = 2$

Energy of the emitted photon:

$$E_4 - E_2 = -0.85 - (-3.40) = 2.55 \text{ eV}$$



Emission & Absorption – Exercise

A hypothetical atom has three energy levels; the ground level and $1.00\ \mathrm{eV}$ and $3.00\ \mathrm{eV}$ above the ground level.

a) Find the frequencies and wavelengths of the spectrum lines that this atom can emit when excited.

b) What wavelengths can be absorbed by this atom if it is initially in its ground level?

Emission & Absorption – Exercise

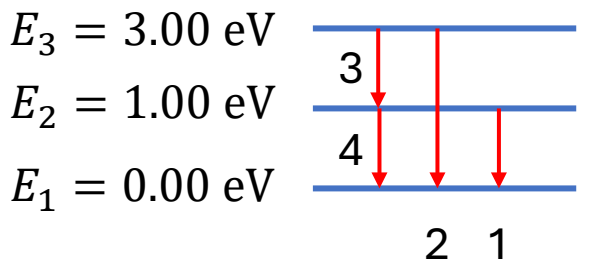
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a) Find the frequencies and wavelengths of the spectrum lines that this atom can emit when excited.

b) What wavelengths can be absorbed by this atom if it is initially in its ground level?

$$\lambda = \frac{c}{f} \qquad \text{and} \qquad hf = E_i - E_f$$

$$\lambda = \frac{hc}{E_i - E_f}$$



$$\lambda_1 = \frac{1240\ \mathrm{eV{\cdot}nm}}{(1-0)\ \mathrm{eV}} = 1240\ \mathrm{nm}, \lambda_2 = \frac{1240\ \mathrm{eV{\cdot}nm}}{(3-0)\ \mathrm{eV}} = 413.3\ \mathrm{nm}$$

$$\lambda_3 = \frac{1240\ \mathrm{eV{\cdot}nm}}{(3-1)\ \mathrm{eV}} = 620\ \mathrm{nm}, \lambda_4 = \frac{1240\ \mathrm{eV{\cdot}nm}}{(1-0)\ \mathrm{eV}} = 1240\ \mathrm{nm}$$

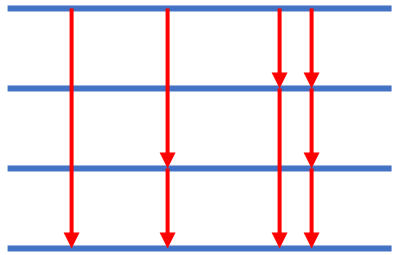
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Emission & Absorption – Exercise

9000 hydrogen atoms are excited and are in an energy state characterized by quantum number $n = 4$. By assuming equal probability for all transitions to lower states, and by following all transitions down to the ground state for all atoms, calculate the total number of photons emitted.

Emission & Absorption – Exercise

9000 hydrogen atoms are excited and are in an energy state characterized by quantum number $n = 4$. By assuming equal probability for all transitions to lower states, and by following all transitions down to the ground state for all atoms, calculate the total number of photons emitted.



9000 / 3 x 1 = 3000
9000 / 3 x 2 = 6000
9000 / 3 / 2 x 3 = 4500
9000 / 3 / 2 x 2 = 3000

16500

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Atoms with $Z > 1$

- Bohr's model works for any atom with a single electron
- Consider an ionized atom with $Z > 1$:

$$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1||q_2|}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r^2}$$

- Here we have Ze^2 instead of e^2, which can be used to derive:

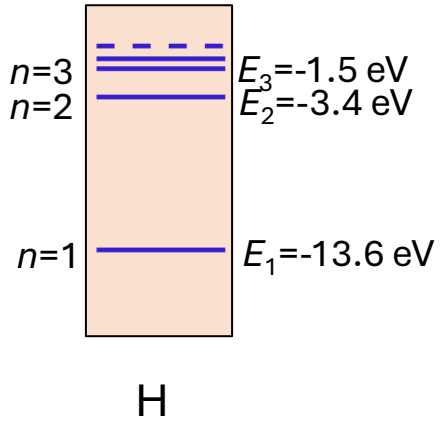
$$r_n = \frac{4\pi\epsilon_0\hbar^2}{Zm_e e^2}\, n^2 = \frac{a_0 n^2}{Z}$$

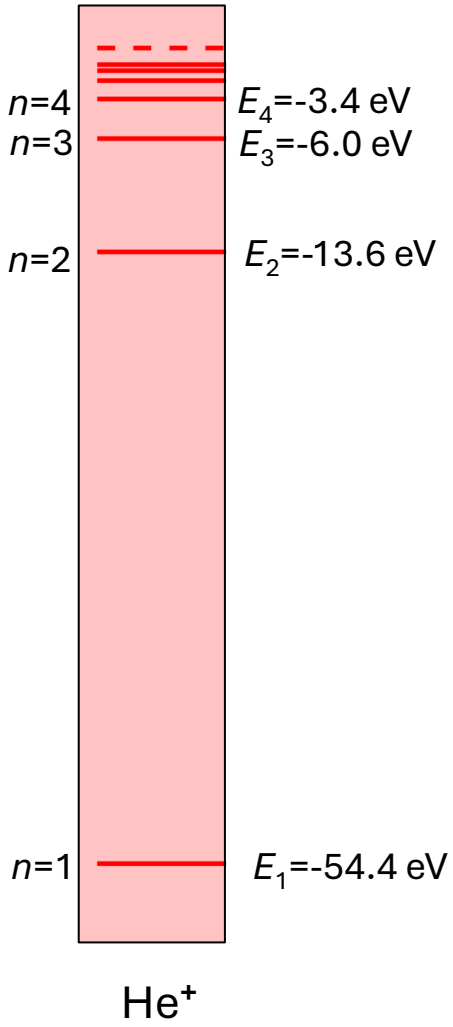
$$E_n = -\frac{m_e(Ze^2)^2}{32\pi^2\epsilon_0^2\hbar^2}\frac{1}{n^2} = -(13.60\ \text{eV})\frac{Z^2}{n^2}$$

Atoms with $Z > 1$

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{Zm_e e^2}\, n^2 = \frac{a_0 n^2}{Z}$$

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Atoms with $Z > 1$ – Exercise

Calculate the two shortest wavelengths of the Balmer series of triply ionized beryllium ($Z = 4$)

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Remember:
Balmer series is to level $n = 2$

Atoms with $Z > 1$ – Exercise

Calculate the two shortest wavelengths of the Balmer series of triply ionized beryllium ($Z = 4$)

$$E_n = -\frac{m_e (Ze^2)^2}{32\pi^2 \epsilon_0^2 \hbar^2} \frac{1}{n^2} = -(13.60\ \text{eV}) \frac{Z^2}{n^2}$$

Remember:
Balmer series is to level $n = 2$

Longest wavelength correspond to $n = 3 \rightarrow n = 2$ and $n = 4 \rightarrow n = 2$

$$E_4 - E_2 = -(13.60\ \text{eV}) 4^2 \left(\frac{1}{4^2} - \frac{1}{2^2} \right) = 40.8\ \text{eV}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\ \text{eV} \cdot \text{nm}}{40.8\ \text{eV}} = 30.4\ \text{nm}$$

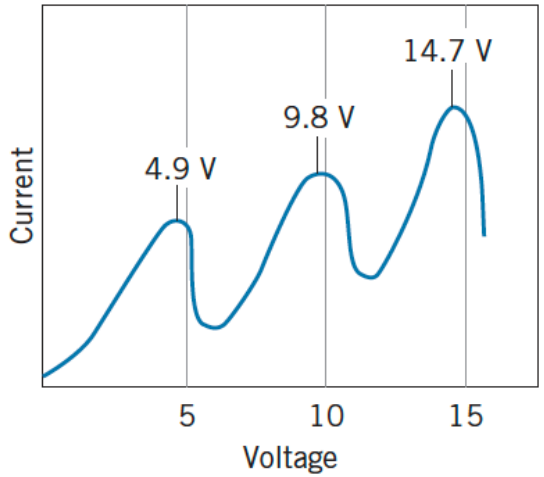
$$E_3 - E_2 = -(13.60\ \text{eV}) 3^2 \left(\frac{1}{3^2} - \frac{1}{2^2} \right) = 30.2\ \text{eV}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{1240\ \text{eV} \cdot \text{nm}}{30.2\ \text{eV}} = 41.0\ \text{nm}$$

The Franck-Hertz Experiment

What happens?

1. Electrons leave (heated) cathode (C) by thermic emission
2. Electrons are attracted by postive grid (G)
3. Electrons collide with atoms (e.g. Hg-atoms) inside the tube
4. Elastic collisions with much heavier Hg-atoms: no energy loss
5. If potential $V > 4.9$ V: electrons have enough E_{kin} to raise the energy level of the Hg-atom → inelastic collision
6. The current drops at multiples of 4.9 V



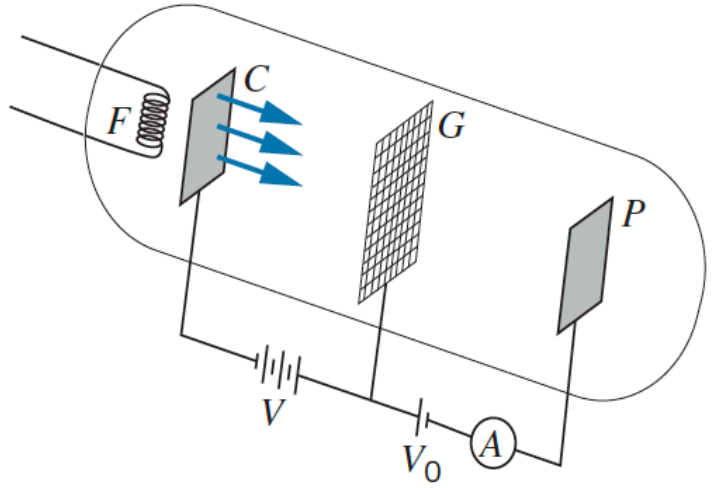


FIGURE 6.21 Franck-Hertz apparatus. Electrons leave the cathode C, are accelerated by the voltage V toward the grid G, and reach the plate P where they are recorded on the ammeter A.

The Franck-Hertz Experiment



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Limits of the Bohr Model

1. Electron moves not around a stationary nucleus, but the nucleus and electron move around their common center of mass
   - H-spectrum: λ increases with a factor 1.00055

2. Wavelength measurements are done in air: $\lambda = \frac{v_{air}}{f}$ instead of $\lambda = \frac{c}{f}$
   - λ decreases with a factor 1.00029

3. Bohr model: smallest angular momentum of electron in hydrogen $L = \hbar$; Experiment gives: $L = 0$.

4. Violates the uncertainty principle. We assume all is in the xy-plane, but this results in $z = 0$ and $\Delta z = 0$, not $\Delta z \Delta p_Z \geq \hbar$